

Spectrum

Let $\mathbf{k}$ be a spherically complete non-archimedean field which is algebraically closed and $A = \mathbf{k}[T]$. We want to consider the “analytic structure” on $\mathbf{mSpec} A$. However, unlike the complex case, the set $\mathbf{mSpec} A$ is totally disconnected with respect to the topology induced by the absolute value on $\mathbf{k}$ (Corollary 28). To overcome this difficulty, Berkovich uses multiplicative semi-norms to “fill in the gaps” between the points in $\mathbf{mSpec} A$, leading to the notion of the spectrum of a Banach ring.

1 Definition and examples

Definition 1. Let R be a Banach ring. The *Berkovich spectrum* $\mathcal{M}(R)$ of R is defined as the set of all multiplicative semi-norms on R that are bounded with respect to the given norm on R . For every point $x \in \mathcal{M}(R)$, we denote the corresponding multiplicative semi-norm by $|\cdot|_x$.

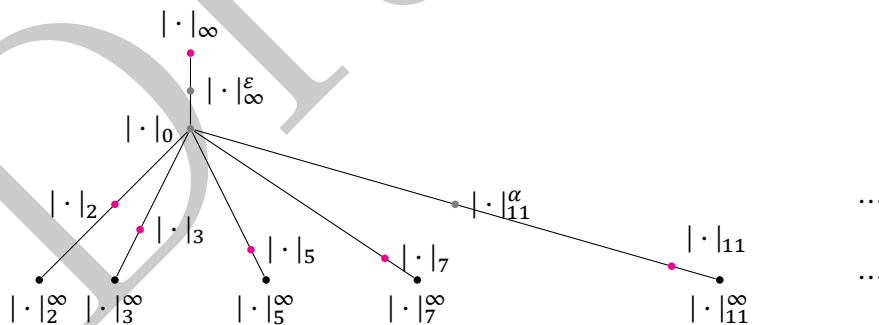
We equip $\mathcal{M}(R)$ with the weakest topology such that for each $f \in R$, the evaluation map $\mathcal{M}(R) \rightarrow \mathbb{R}_{\geq 0}$, defined by $x \mapsto |f|_x =: f(x)$, is continuous.

Example 2. Let $(\mathbf{k}, |\cdot|)$ be a complete valuation field. The Berkovich spectrum $\mathcal{M}(\mathbf{k})$ consists of a single point corresponding to the given absolute value $|\cdot|$ on $\mathbf{k}$.

Example 3. Consider the Banach ring $(\mathbb{Z}, \|\cdot\|)$ with $\|\cdot\| = |\cdot|_\infty$ is the usual absolute value norm on $\mathbb{Z}$. Let $|\cdot|_p$ denote the p -adic norm for each prime number p , i.e., $|n|_p = p^{-v_p(n)}$ for each $n \in \mathbb{Z}$, where $v_p(n)$ is the p -adic valuation of n . The Berkovich spectrum

$$\mathcal{M}(\mathbb{Z}) = \{|\cdot|_\infty^\varepsilon : \varepsilon \in (0, 1]\} \cup \{|\cdot|_p^\alpha : p \text{ is prime}, \alpha \in (0, \infty]\} \cup \{|\cdot|_0\},$$

where $|a|_p^\infty := \lim_{\alpha \rightarrow \infty} |a|_p^\alpha$ for each $a \in \mathbb{Z}$ and $|\cdot|_0$ is the trivial norm on $\mathbb{Z}$.



Yang: To be continued.

Theorem 4. Let R be a Banach ring. The spectrum $\mathcal{M}(R)$ is nonempty.

Proof. Let $\mathfrak{m}$ be a maximal ideal of R . Note that the pullback of residue norm on the residue field $R/\mathfrak{m}$ is bounded with respect to the given norm on R . Replacing R by the completion of $R/\mathfrak{m}$, we may assume that R is a complete field. Consider the set

$$\Sigma = \{\text{norm on } R \text{ bounded by the given norm } \|\cdot\|\}$$

with the partial order defined by boundedness. Since for a descending chain in Σ , the infimum is a norm, by Zorn's lemma, there exists a minimal element $|\cdot| \in \Sigma$.

We claim that $|\cdot|$ is multiplicative. Since the spectral radius $\rho(f) = \lim_{n \rightarrow \infty} |f^n|^{1/n}$ associated to $|\cdot|$ is power-multiplicative and bounded by $|\cdot|$, by minimality of $|\cdot|$, we have $\rho(f) = |f|$ for each $f \in R$. Thus $|\cdot|$ is power-multiplicative. If $|\cdot|$ is not multiplicative, then there exist $a, b \in R \setminus \{0\}$ such that $|ab| < |a||b|$. Then $|b| \leq |a^{-1}||ab| < |a^{-1}||a||b|$, which implies that $|a||a^{-1}| > 1$. Set $r = |a|^{-1} < |a^{-1}|$ and consider $R\langle T/r \rangle$. Since $r \cdot |a| = 1$, we have that

$$\left| \sum_{n=0}^{\infty} a^n T^n \right| = \sum_{n=0}^{\infty} |a^n| r^n = \sum_{n=0}^{\infty} |a|^n r^n = \sum_{n=0}^{\infty} 1 = \infty.$$

The power series is not convergent in $R\langle T/r \rangle$ and hence $1 - aT$ is not invertible in $R\langle T/r \rangle$. Let $\mathfrak{n}$ be a maximal ideal of $R\langle T/r \rangle$ containing $1 - aT$. Consider $R \rightarrow R\langle T/r \rangle \rightarrow R\langle T/r \rangle / \mathfrak{n}$. Since R is a field, the composition is injective. The residue norm on $R\langle T/r \rangle / \mathfrak{n}$ induces a norm $|\cdot|'$ on R bounded by $|\cdot|$. Note that $|a^{-1}|' \leq |T| = r = |a|^{-1} < |a^{-1}|$, contradicting the minimality of $|\cdot|$. $\square$

Definition 5. Let $\varphi : R \rightarrow S$ be a bounded ring homomorphism of Banach rings. The *pullback* map $\mathcal{M}(\varphi) : \mathcal{M}(S) \rightarrow \mathcal{M}(R)$ is defined by $\mathcal{M}(\varphi)(x) = x \circ \varphi : f \mapsto |\varphi(f)|_x$ for each $x \in \mathcal{M}(S)$.

Note that $\mathcal{M}(\varphi)(f^{-1}(V)) = \varphi(f)^{-1}(V)$ for each $f \in R$ and open subset $V \subset \mathbb{R}_{\geq 0}$. Hence the pullback map is continuous.

Notation 6. Let R be a Banach ring and $x \in \mathcal{M}(R)$. We denote by $|\cdot|_x$ the multiplicative semi-norm on R corresponding to the point x . Its kernel $\{f \in R : |f|_x = 0\}$ is a closed prime ideal of R , denoted by $\wp_x$.

Definition 7. Let R be a Banach ring. For each $x \in \mathcal{M}(R)$, the *completed residue field* at the point x is defined as the completion of the residue field $\kappa(x) = \text{Frac}(R/\wp_x)$ with respect to the multiplicative norm induced by the semi-norm $|\cdot|_x$, denoted by $\mathcal{H}(x)$.

Example 8. Consider the Banach ring $(\mathbb{Z}, |\cdot|_{\infty})$ as in [Example 3](#). We have

- $x = |\cdot|_{\infty}^{\varepsilon}$ for some $\varepsilon \in (0, 1]$: $\wp_x = (0)$ and $\mathcal{H}(x) \cong \mathbb{R}$ with the absolute value norm raised to the power ε ;
- $x = |\cdot|_0$: $\wp_x = (0)$ and $\mathcal{H}(x) \cong \mathbb{Q}$ with the trivial norm;
- $x = |\cdot|_p^{\alpha}$ for some prime number p and $\alpha \in (0, \infty)$: $\wp_x = (0)$ and $\mathcal{H}(x) \cong \mathbb{Q}_p$ with the p -adic norm raised to the power α ;
- $x = |\cdot|_p^{\infty}$ for some prime number p : $\wp_x = (p)$ and $\mathcal{H}(x) \cong \mathbb{F}_p$ with the trivial norm.

Definition 9. Let R be a Banach ring. The *Gel'fand transform* of R is the bounded ring homomorphism

$$\Gamma : R \rightarrow \prod_{x \in \mathcal{M}(R)} \mathcal{H}(x), \quad f \mapsto (f(x))_{x \in \mathcal{M}(R)},$$

where the norm on the product $\prod_{x \in \mathcal{M}(R)} \mathcal{H}(x)$ is given by the supremum norm.

Example 10. Let K be a number field equipped with [Yang: To be continued](#).

Proposition 11. The Gel'fand transform $\Gamma : R \rightarrow \prod_{x \in \mathcal{M}(R)} \mathcal{H}(x)$ of a Banach ring R factors through the uniformization R^u of R , and the induced map $R^u \rightarrow \prod_{x \in \mathcal{M}(R)} \mathcal{H}(x)$ is an isometric embedding. Yang: To be checked.

Proof. Yang: To be added. □

Lemma 12. Let $\{K_i\}_{i \in I}$ be a family of completed fields. Consider the Banach ring $R = \prod_{i \in I} K_i$ equipped with the product norm. The spectrum $\mathcal{M}(R)$ is homeomorphic to the Stone-Čech compactification of the discrete space I .

Proof. Yang: To be added. □

Remark 13. Here we list some facts about ultrafilters and the Stone-Čech compactification that will be used in the proof of Lemma 12.

Let I be a set. A *filter* on I is a nonempty collection $\mathcal{F} \subset \mathcal{P}(I)$ of subsets of I such that:

- if $A, B \in \mathcal{F}$, then $A \cap B \in \mathcal{F}$;
- if $A \in \mathcal{F}$ and $A \subset B \subset I$, then $B \in \mathcal{F}$.

For example, let $I = \mathbb{N}$ and

$$\mathcal{F} = \{A \subset \mathbb{N} : \#(\mathbb{N} \setminus A) < \infty\}.$$

Then $\mathcal{F}$ is a filter on $\mathbb{N}$, called the *Fréchet filter*. Another example is the filter called the *principal filter* generated by a point $i_0 \in I$:

$$\mathcal{F}_{i_0} = \{A \subset I : i_0 \in A\}.$$

Filters are partially ordered by inclusion. A filter $\mathcal{U}$ is called an *ultrafilter* if it is maximal with respect to inclusion. A filter is an ultrafilter if and only if for every subset $A \subset I$, either $A \in \mathcal{U}$ or $I \setminus A \in \mathcal{U}$.

One can define the notion of convergence along ultrafilters in a topological space. Let X be a topological space, $(x_i)_{i \in I}$ a collection of points in X indexed by a set I , and $\mathcal{U}$ an ultrafilter on I . We say that $(x_i)_{i \in I}$ *converges* to a point $x \in X$ along the ultrafilter $\mathcal{U}$ if for every open neighborhood U of x , the set $\{i \in I : x_i \in U\}$ belongs to $\mathcal{U}$. It is denoted by $\lim_{\mathcal{U}} x_i = x$. When $I = \mathbb{N}$ and $\mathcal{U}$ is the Fréchet filter, this notion of convergence coincides with the usual notion of convergence of sequences.

The Stone-Čech compactification of a discrete space is the largest compact Hausdorff space in which the original space can be densely embedded. Yang: To be checked.

Theorem 14. Let R be a Banach ring. The spectrum $\mathcal{M}(R)$ is a compact Hausdorff space.

Proof. Yang: To be added. □

Corollary 15. Let R be a Banach ring and R^u its uniformization. The canonical map $\mathcal{M}(R^u) \rightarrow \mathcal{M}(R)$ induced by the bounded homomorphism $R \rightarrow R^u$ is a homeomorphism.

Proposition 16. Let $\mathbf{l}/\mathbf{k}$ be a finite normal extension of complete valuation fields and A a Banach $\mathbf{k}$ -algebra. Then we have:

- (a) if $\mathbf{l}/\mathbf{k}$ is Galois, then $\mathcal{M}(A \otimes_{\mathbf{k}} \mathbf{l})$ admits a natural action of the Galois group $\text{Gal}(\mathbf{l}/\mathbf{k})$ and the quotient space is homeomorphic to $\mathcal{M}(A)$;
- (b) if $\mathbf{l}/\mathbf{k}$ is purely inseparable, then the canonical map $\mathcal{M}(A \otimes_{\mathbf{k}} \mathbf{l}) \rightarrow \mathcal{M}(A)$ is a homeomorphism.

Proof. Yang: To be added. □

Corollary 17. Let $\mathbf{k}$ be a complete valuation field, $\mathbf{k}$ its completion of an algebraic closure, and A a Banach $\mathbf{k}$ -algebra. Then we have a natural homeomorphism $\mathcal{M}(A \hat{\otimes}_{\mathbf{k}} \mathbf{k}) / \text{Gal}(\mathbf{k}/\mathbf{k}) \cong \mathcal{M}(A)$.

Proof. Yang: To be added. □

2 Reduction map and kernel map

Proposition 18. Let R be a Banach ring. The kernel map $\mathcal{M}(R) \rightarrow \text{Spec}(R), x \mapsto \wp_x$ is continuous with respect to the Zariski topology on $\text{Spec}(R)$.

Proof. Let $D(f) = \{f \neq 0\} \subset \text{Spec}(R)$ be a principal open subset for some $f \in R$. The preimage of $D(f)$ under the kernel map is just the set $\{x \in \mathcal{M}(R) : |f|_x > 0\} = f^{-1}(\mathbb{R}_{>0})$, which is open in $\mathcal{M}(R)$ by definition of the topology on $\mathcal{M}(R)$. □

Example 19. Let us consider the spectrum $\mathcal{M}(\mathbb{Z})$ in Example 3. Under the kernel map $\mathcal{M}(\mathbb{Z}) \rightarrow \text{Spec}(\mathbb{Z})$, the points $|\cdot|_p^\infty$ for each prime number p are mapped to the prime ideal (p) , the other above points are all mapped to the zero ideal (0) .

Proposition 20. Let A be an $\mathbf{k}$ -affinoid algebra over a complete non-archimedean field $\mathbf{k}$. Then the kernel map $\mathcal{M}(A) \rightarrow \text{Spec}(A)$ is surjective.

Proof. Yang: To be add. □

Construction 21. Suppose that R is a non-archimedean Banach ring with valuation subring R° and maximal ideal $R^{\circ\circ}$. For each $x \in \mathcal{M}(R)$, there is an induced homomorphism $R^\circ \rightarrow \mathcal{H}(x)^\circ$ between the valuation subrings. Furthermore, we have an induced homomorphism between the residue rings $\tilde{R} = R^\circ / R^{\circ\circ} \rightarrow \mathcal{K}_{\mathcal{H}(x)}$. This gives rise to the *reduction map*

$$\text{Red} : \mathcal{M}(R) \rightarrow \text{Spec}(\tilde{R}), \quad x \mapsto \ker(\tilde{R} \rightarrow \mathcal{K}_{\mathcal{H}(x)}).$$

Let $f : R \rightarrow S$ be a bounded homomorphism of non-archimedean Banach rings. Yang: The following diagram commutes:

$$\begin{array}{ccc} \mathcal{M}(S) & \xrightarrow{\text{Red}} & \text{Spec}(\tilde{S}) \\ \mathcal{M}(f) \downarrow & & \downarrow \text{Spec}(\tilde{f}) \\ \mathcal{M}(R) & \xrightarrow{\text{Red}} & \text{Spec}(\tilde{R}) \end{array}$$

Hence the reduction map is functorial with respect to bounded homomorphisms of non-archimedean Banach rings.

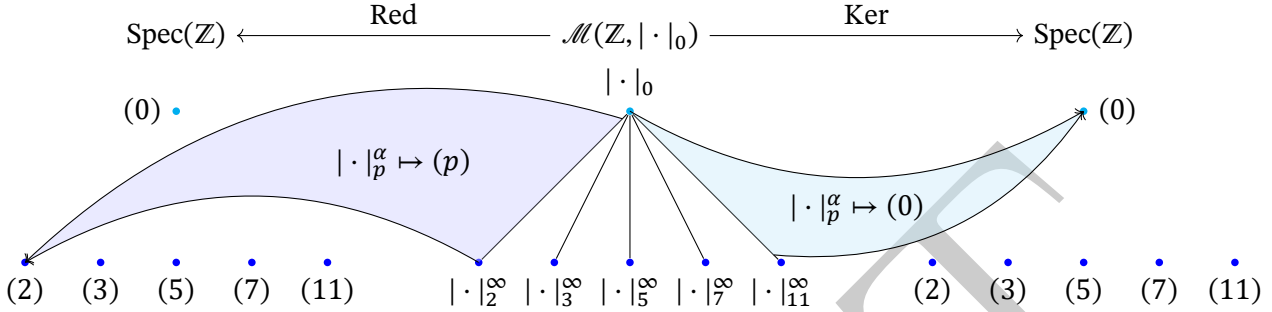
Example 22. Let $(\mathbb{Z}, |\cdot|_0)$ be the Banach ring with the trivial norm. The reduction ring is $\tilde{\mathbb{Z}} = \mathbb{Z}$.

- $x = |\cdot|_p^\alpha$ for some prime number p and $\alpha \in (0, \infty]$: $\mathcal{K}_{\mathcal{H}(x)} \cong \mathbb{F}_p$ and the induced homomorphism

$\tilde{\mathbb{Z}} = \mathbb{Z} \rightarrow \kappa_{\mathcal{H}(x)} = \mathbb{F}_p$ is the natural projection $\mathbb{Z} \rightarrow \mathbb{F}_p$;

- $x = |\cdot|_0: \kappa_{\mathcal{H}(x)} \cong \mathbb{Q}$ and the induced homomorphism $\tilde{\mathbb{Z}} = \mathbb{Z} \rightarrow \kappa_{\mathcal{H}(x)} = \mathbb{Q}$ is the natural inclusion $\mathbb{Z} \rightarrow \mathbb{Q}$.

The following diagram illustrates the reduction map and the kernel map for the spectrum $\mathcal{M}(\mathbb{Z}, |\cdot|_0)$:



Proposition 23. Let R be a non-archimedean Banach ring and $\tilde{U} \subset \text{Spec}(\tilde{R})$ be a Zariski open subset. Then the preimage $\text{Red}^{-1}(\tilde{U})$ is a closed subset of $\mathcal{M}(R)$.

Proof. Yang: To be completed. □

Proposition 24. Let $\mathbf{k}$ be a complete non-archimedean field and A, B a strictly $\mathbf{k}$ -affinoid algebras. Suppose that $\varphi: A \rightarrow B$ is a bounded $\mathbf{k}$ -algebra homomorphism. Then we have:

- if φ is finite, then the induced homomorphism $\tilde{\varphi}: \tilde{A} \rightarrow \tilde{B}$ is finite;
- if φ is of finite type, then the induced homomorphism $\tilde{\varphi}: \tilde{A} \rightarrow \tilde{B}$ is of finite type.

Yang: To be revised.

3 Spectrum of Tate algebras

In this subsection, fix an algebraically closed complete non-archimedean field $\mathbf{k}$.

Spectrum of Tate algebra in one variable Let $A = \mathbf{k}\{T/r\}$. We list some types of points in the spectrum $\mathcal{M}(A)$.

Construction 25. For each $a \in \mathbf{k}$ with $|a| \leq r$, we have the *type I* point x_a corresponding to the evaluation at a , i.e., $|f|_{x_a} := |f(a)|$ for each $f \in A$.

For each closed disk $E = E(a, s) := \{b \in \mathbf{k} : |b - a| \leq s\}$ with center $a \in \mathbf{k}$ and radius $s \leq r$, we have the point $x_E = x_{a,s}$ corresponding to the multiplicative semi-norm defined by

$$|f|_{x_E} = |f|_{x_{a,s}} := \sup_{b \in E(a,s)} |f(b)|$$

for each $f \in A$. If $s \in |\mathbf{k}^\times|$, then the point x_E is called a *type II* point; otherwise, it is called a *type III* point.

Let $E_n = E(a_n, s_n)$ be a sequence of closed disks in $\mathbf{k}$ such that $E_{n+1} \subsetneq E_n$ and $\bigcap_n E_n = \emptyset$. Then we have the point $x_{\{E_n\}} = x_{\{a_n, s_n\}}$ corresponding to the multiplicative semi-norm defined by

$$|f|_{x_{\{E_n\}}} = |f|_{x_{\{a_n, s_n\}}} := \inf_n |f|_{x_{E_n}}$$

for each $f \in A$. Such a point is called a *type IV* point.

Proposition 26. The points in the spectrum $\mathcal{M}(\mathbb{k}\{r^{-1}T\})$ can be classified into four types as described in [Construction 25](#).

Proof. Fix $x \in \mathcal{M}(\mathbb{k}\{r^{-1}T\})$, set

$$s = \inf_{a \in \mathbb{k}} |T - a|_x \leq r, \quad E = \{a \in \mathbb{k} : |T - a|_x = s\} \subset E(0, r).$$

Case 1. $E \neq \emptyset$ and $s = 0$.

By assumption, there exists $a \in E$ such that $|T - a|_x = 0$. Note that if $f(a) = 0$, then $T - a \mid f$ in $\mathbb{k}\{r^{-1}T\}$ and hence $|f|_x = |T - a|_x |g|_x = 0$. Then we have

$$|f(a)| = ||f(a)| - |f(a) - f|_x| \leq |f|_x \leq |f(a)| + |f(a) - f|_x = |f(a)|$$

for each $f \in \mathbb{k}\{r^{-1}T\}$, which implies that $|f|_x = |f(a)|$. Thus x is a type I point x_a .

Case 2. $E \neq \emptyset$ and $s > 0$.

Let $a \in E$. Note that for every $b \in \mathbb{k}$, we have

$$|a - b| \leq \max\{|T - a|_x, |T - b|_x\} = |T - b|_x.$$

First we show that $E = E(a, s)$. For each $b \in E(a, s)$, we have $|T - b|_x \leq \max\{|T - a|_x, |a - b|\} = s$, which implies that $b \in E$. Conversely, for each $b \in E$, we have $|a - b| \leq \max\{|T - a|_x, |T - b|_x\} = s$.

Let $f \in \mathbb{k}[T]$ be a polynomial. Write $f = \prod_{i=1}^n (T - c_i)$ for some $c_1, \dots, c_n \in \mathbb{k}$. Then we have

$$|f|_x = \prod_{i=1}^n |T - c_i|_x \geq \prod_{i=1}^n |b - c_i| = |f(b)|, \quad \forall b \in E.$$

I claim that for every $\varepsilon \in (0, 1)$, there exists $b \in E$ such that $\varepsilon |T - c_i|_x < |b - c_i|$ for each $i = 1, \dots, n$. Indeed, if $c_i \notin E$, then $|T - c_i|_x = |b - c_i|$ for each $b \in E$. Hence we only need to consider the case when $c_i \in E$. Since $\mathbb{k}$ is algebraically closed, $E(a, s) \setminus \bigcup_{i=1}^n E(c_i, \varepsilon s) \neq \emptyset$. Choose b in the set. Then we have

$$|f(b)| \geq \prod_{i=1}^n \varepsilon |T - c_i|_x = \varepsilon^n |f|_x.$$

Thus $|f|_x = \sup_{b \in E} |f(b)|$ for each polynomial $f \in \mathbb{k}[T]$. Since polynomials are dense in $\mathbb{k}\{r^{-1}T\}$, we have $|f|_x = \sup_{b \in E} |f(b)|$ for each $f \in \mathbb{k}\{r^{-1}T\}$. Therefore, x is the point $x_E = x_{a,s}$, which is of type II or type III depending on whether $s \in |\mathbb{k}^\times|$ or not.

Case 3. $E = \emptyset$.

Set $E_n = \{a \in \mathbb{k} : |T - a|_x \leq s + 1/n\}$ and $a_n \in E_n$ for each $n \in \mathbb{N}$. By the similar argument as in [Case 2](#), we have $E_n = E(a_n, s + 1/n)$. Note that E_n is a decreasing sequence of closed disks with $\bigcap_n E_n = E = \emptyset$.

For $c \in \mathbb{k}$, there exists N such that $\forall n \geq N$, we have

$$c \notin E_n \implies |T - c|_x > |T - a_n|_x \implies |T - c|_x = |a_n - c|.$$

Thus

$$\inf_n |T - c|_{E_n} = \inf_n |a_n - c| = |T - c|_x.$$

By multiplicativity, we have $\inf_n |f|_{E_n} = |f|_x$ for each polynomial $f \in \mathbb{k}[T]$. And then by density of polynomials, the equality holds for each $f \in \mathbb{k}\{r^{-1}T\}$. Therefore, $x = x_{\{E_n\}} = x_{\{a_n, s_n\}}$ is of type IV. $\square$

Now we describe the structure of $X = \mathcal{M}(\mathbb{k}\{T\})$ in more details. For each $a \in \mathbb{k}$ with $|a| \leq 1$, there is a line segment $L_{\xi, x_a} = \{x_{a,r} | r \in [0, 1]\}$ in X connecting the Gauss point $x_{0,1}$ and the type I point x_a . Two such line segments L_{ξ, x_a} and L_{ξ, x_b} will meet at the point $x_{a,s} = x_{b,s}$ where $s = |a - b|$.

On such a line segment L_{ξ, x_a} , if $s \in |\mathbb{k}^\times|$, then the point $x_{a,s}$ is of type II; otherwise, it is of type III. One can think of the type II points like rational points and the type III points like irrational points on the line segment.

On a type II point $x_{a,s}$, there are infinitely many line segments $L_{x_{a,s}, x_b}$ connecting $x_{a,s}$ and type I points x_b with $|b - a| < s$. It looks like a tree with infinitely many branches rooted at $x_{a,s}$. The branch is indexed by the quotient set $E(a, s)/B(a, s) \cong \mathbb{k}_k$. On a type III point $x_{a,s}$, there are no line segments branching out. In a slogan, the spectrum $\mathcal{M}(\mathbb{k}\{T\})$ looks like a fractal tree with infinitely many branches. In the tree, $|\mathbb{k}^\times|$ controls the placement of roots of branches, while $\mathbb{k}_k$ controls the number of branches at each root.

Consider a sequence of disks $E_n = E(a_n, s_n)$ such that $E_{n+1} \subset E_n$. It corresponds to a piecewise linear path $L_{x_{a_1, s_1}, x_{a_2, s_2}} \cup L_{x_{a_2, s_2}, x_{a_3, s_3}} \cup \dots$ in the tree. Topologically, it is homeomorphic to $[0, 1)$. If the intersection $\bigcap_n E_n$ is nonempty, then it corresponds to a type I, II, or III point fitting into the place of 1 in the path. It is compact and everything is fine. However, if the intersection $\bigcap_n E_n$ is empty, then it will look like $[0, 1)$ with a missing point 1 locally if we only consider the points of type I, II, and III. The type IV point $x_{\{E_n\}}$ is exactly the missing point 1 in the path, which makes the path compact.

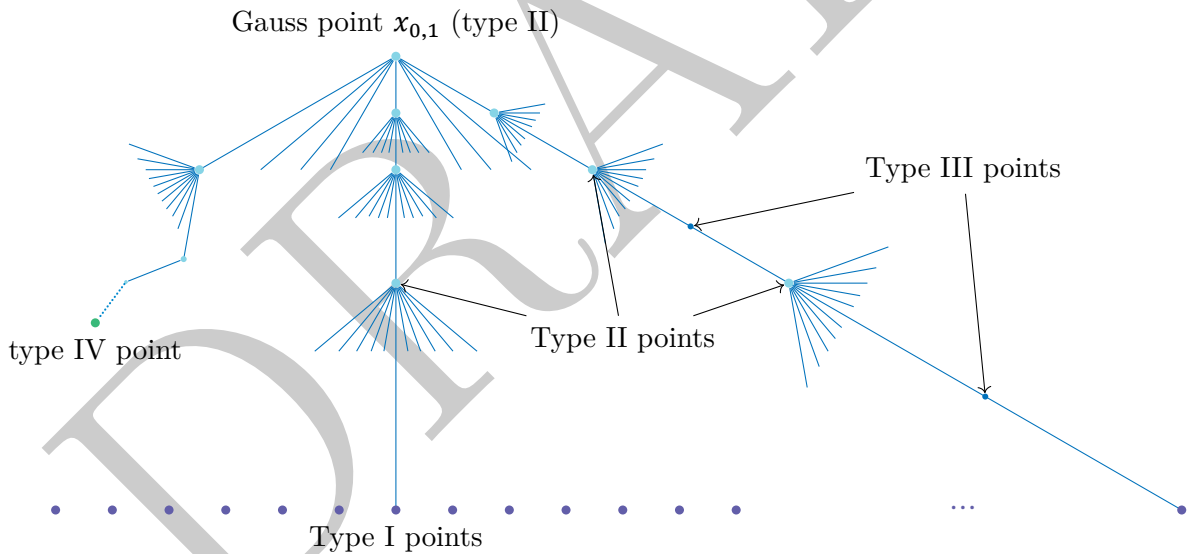


Figure 1: The Berkovich spectrum $\mathcal{M}(\mathbb{k}\{T\})$

Kernel map for $\mathcal{M}(\mathbb{k}\{T/r\})$ If $x = x_a$ is a type I point corresponding to $a \in \mathbb{k}$ with $|a| \leq r$, then $\wp_{x_a} = (T - a)$. For type II, type III or type IV points x , by Yang: ref, we only need to consider elements in A of the form $T - c$ for some $c \in \mathbb{k}$ with $|c| \leq r$. Yang: Since $|T - c|_x \neq 0$ for all $c \in \mathbb{k}$ with $|c| \leq r$, we have $\wp_x = (0)$. To summarize, the kernel map $\mathcal{M}(A) \rightarrow \text{Spec}(A)$ is given by

$$\wp_x = \begin{cases} (T - a), & \text{if } x = x_a \text{ is a type I point;} \\ (0), & \text{if } x \text{ is a type II, type III or type IV point.} \end{cases}$$

Proposition 27. The completed residue fields of the four types of points in the spectrum $\mathcal{M}(\mathbb{k}\{r^{-1}T\})$ are described as follows:

- type I point x_a : $\mathcal{H}(x_a) \cong \mathbb{k}$;
- type II point $x_{a,s}$: $\mathcal{H}(x_{a,s}) \cong \widehat{\text{Frac}}(\mathbb{k}\{S\})$ with $S = (T - a)/s$;
- type III point $x_{a,s}$: $\mathcal{H}(x_{a,s}) \cong \mathbf{L}_s(\mathbb{k}) = \{f = \sum_{n \in \mathbb{Z}} a_n S^n : \lim_{|n| \rightarrow \infty} |a_n| s^n = 0\}$;
- type IV point $x_{\{a_n, s_n\}}$: $\mathcal{H}(x_{\{a_n, s_n\}})$ is an immediate extension of $\mathbb{k}$.

In particular, for a type II point $x_{a,s}$, $\mathcal{H}(x_{a,s}) \cong \mathcal{H}_{\mathbb{k}}(S)$ and $|\mathcal{H}(x_{a,s})^\times| = |\mathbb{k}^\times|$; for a type III point $x_{a,s}$, $\mathcal{H}(x_{a,s}) \cong \mathcal{H}_{\mathbb{k}}$ and $|\mathcal{H}(x_{a,s})^\times| = |\mathbb{k}^\times| \cdot s^{\mathbb{Z}}$.

Proof. Let $x = x_a$ be a type I point. Then $\wp_x = (T - a)$ and hence $\mathbb{k}\{r^{-1}T\}/\wp_x \cong \mathbb{k}$ as $\mathbb{k}$ -algebras. For every $f \in \mathbb{k}\{r^{-1}T\}$, we have $|f + (T - a)g|_x = |f(a)|$ for each $g \in \mathbb{k}\{r^{-1}T\}$. Thus the norm on $\mathbb{k}\{r^{-1}T\}/\wp_x$ induced by the point x coincides with the norm on $\mathbb{k}$. Therefore, $\mathcal{H}(x_a) \cong \mathbb{k}$.

Let $x = x_{a,s}$ be a type II point. Set $S = (T - a)/s$. Then there is an injective isometric $\mathbb{k}$ -algebra homomorphism $(\mathbb{k}\{T/r\}, |\cdot|_x) \rightarrow \mathbb{k}\{S\}$ sending T to $a + sS$. This is well-defined since $|a|, s \leq r$ and hence $|a + sS| \leq r$. Note that it may not be surjective. Natively you can image it is given by restricting the domain of functions in $\mathbb{k}\{T/r\}$ from $E(0, r)$ to $E(a, s)$. It has dense image since $\mathbb{k}[S] \subset \mathbb{k}\{S\}$ is dense. By injectivity, we have $\widehat{\text{Frac}}((\mathbb{k}\{T/r\}, |\cdot|_x)) \subset \widehat{\text{Frac}}(\mathbb{k}\{S\})$. By density of the image, we have $\widehat{\text{Frac}}((\mathbb{k}\{T/r\}, |\cdot|_x)) = \widehat{\text{Frac}}(\mathbb{k}\{S\})$. Hence we have $\mathcal{H}(x_{a,s}) \cong \widehat{\text{Frac}}(\mathbb{k}\{S\})$.

To show the assertion of values groups and residue fields, we only need to consider $\text{Frac}(\mathbb{k}\{S\})$. Since $|\mathbb{k}\{S\}| = |\mathbb{k}|$, we have $|\text{Frac}(\mathbb{k}\{S\})^\times| = |\mathbb{k}^\times|$. Every element in $\text{Frac}(\mathbb{k}\{S\})^\circ$ can be written as $(f + \varepsilon)/(g + \delta)$ for some $f, g \in \mathbb{k}^\circ[S]$ with $|\varepsilon| < |f|$, $|\delta| < |g|$ and $|f| \leq |g|$. After reduction, we have $(f + \varepsilon)/(g + \delta) = \tilde{f}/\tilde{g}$ with $\tilde{f}, \tilde{g} \in \mathcal{H}_{\mathbb{k}}[S]$. It follows that $\mathcal{H}_{\mathcal{H}(x_{a,s})} \subset \mathcal{H}_{\mathbb{k}}(S)$. The other inclusion is clear since $\mathcal{H}_{\mathbb{k}}[S] \subset \mathbb{k}\{S\}^\circ$. Then we get $\mathcal{H}_{\mathcal{H}(x_{a,s})} \cong \mathcal{H}_{\mathbb{k}}(S)$.

Let $x = x_{a,s}$ be a type III point. By [Yang: ref](#), $\mathbf{L}_s(\mathbb{k})$ is indeed a complete non-archimedean field with respect to the norm $|f|_{\mathbf{L}} = \sup_{n \in \mathbb{Z}} |a_n| s^n$. Denote by S the indeterminate of $\mathbf{L}_s(\mathbb{k})$. We have an injective isometric $\mathbb{k}$ -algebra homomorphism $(\mathbb{k}\{T/r\}, |\cdot|_x) \rightarrow \mathbf{L}_s(\mathbb{k})$ sending T to $a + S$. Easily see that $\mathbb{k}[S, S^{-1}] \subset \mathbf{L}_s(\mathbb{k})$ is dense. It follows that $\text{Frac}(\mathbb{k}\{T/r\})$ is dense in $\mathbf{L}_s(\mathbb{k})$. Then we have $\mathcal{H}(x_{a,s}) = \mathbf{L}_s(\mathbb{k})$.

Since $|a_n| s^n \neq 1$ if $n \neq 0$, we have $\mathcal{H}_{\mathcal{H}(x_{a,s})} = \mathcal{H}_{\mathbb{k}}$. For every $f = \sum_{n \in \mathbb{Z}} a_n S^n \in \mathbf{L}_s(\mathbb{k})$, we have $|f|_{\mathbf{L}} = |a_{n_0}| s^{n_0}$ for some $n_0 \in \mathbb{Z}$. Thus $|f|_{\mathbf{L}} \in |\mathbb{k}^\times| \cdot s^{\mathbb{Z}}$. And easily see every element in $|\mathbb{k}^\times| \cdot s^{\mathbb{Z}}$ can be achieved by some $f \in \mathbf{L}_s(\mathbb{k})$.

Let $x = x_{\{a_n, s_n\}}$ be a type IV point.

[Yang: To be completed.](#) □

Reduction map for $\mathcal{M}(\mathbb{k}\{T/r\})$ The reduction ring is

$$\tilde{A} = \begin{cases} \mathcal{H}_{\mathbb{k}}[T], & \text{if } r \in |\mathbb{k}^\times|; \\ \mathcal{H}_{\mathbb{k}}, & \text{if } r \notin |\mathbb{k}^\times|. \end{cases}$$

If $r \notin |\mathbb{k}^\times|$, then the reduction map $\text{Red} : \mathcal{M}(A) \rightarrow \text{Spec}(\tilde{A}) = \text{Spec}(\mathcal{H}_{\mathbb{k}})$ is trivial, i.e., every point in $\mathcal{M}(A)$ is mapped to the zero ideal of $\mathcal{H}_{\mathbb{k}}$. Hence we only consider the case when $r \in |\mathbb{k}^\times|$. Without loss of generality, we may assume that $r = 1$.

For each type I point x_a with $a \in \mathbb{k}$ and $|a| \leq 1$, the induced homomorphism $\tilde{A} = \mathbb{k}_{\mathbb{k}}[T] \rightarrow \mathbb{k}_{\mathcal{H}(x_a)} = \mathbb{k}_{\mathbb{k}}$ is given by $T \mapsto a \pmod{\mathbb{k}^{\circ}}$. Thus the reduction map $\text{Red} : \mathcal{M}(A) \rightarrow \text{Spec}(\tilde{A}) = \text{Spec}(\mathbb{k}_{\mathbb{k}}[T])$ sends the type I point x_a to the maximal ideal $(T - \tilde{a})$. For a type II or type III point $x_{a,s}$ with $a \in \mathbb{k}$ and $s \leq 1$, the homomorphism $A^{\circ} \rightarrow \mathcal{H}(x_{a,s})^{\circ}$

Yang: To be continued.

Spectrum of Tate algebra in several variables Let $A = \mathbb{k}\{T\}$. To consider the spectrum $\mathcal{M}(A)$, we first look at the image of the points in $\mathcal{M}(A)$ under the kernel map $\mathcal{M}(A) \rightarrow \text{Spec}(A)$.

For every $\mathfrak{p} \in \text{Spec}(A)$, since A is noetherian, $\mathfrak{p}$ is finitely generated, say $\mathfrak{p} = (f_1, \dots, f_n)$. On $E(0, 1) \subset \mathbb{k}^n$, $\mathfrak{p}$ defines a closed analytic subspace $V = V(f_1, \dots, f_n) = \{x \in E(0, 1) : f_i(x) = 0, i = 1, \dots, n\}$. By Yang: ref, V is nonempty. On $E(0, 1)$, we have the supremum norm $\|\cdot\|_{\text{sup}}$. It inherits the metric from $E(0, 1)$ and hence is a complete metric space.

According to the proof of Proposition 26, we should use prime divisor on $\text{Spec}(A/\mathfrak{p})$ to replace the role of points in one variable case.

Yang: To be continued.

Appendix

Corollary 28. Let (X, d) be an ultra-metric space. Then X is totally disconnected, i.e., the only connected subsets of X are the set with at most one point.